

Experiment 8: Linear Regression

Aim

To implement the Linear Regression algorithm in Python for predicting continuous values.

Algorithm

Step 1: Import the required libraries.

Step 2: Generate a regression dataset using `make_regression()`.

Step 3: Split the dataset into training and testing sets.

Step 4: Create a Linear Regression model.

Step 5: Train the model using the training data.

Step 6: Predict the output for the test data.

Step 7: Calculate and display the R^2 score of the model.

Program

```
#Experiment 8: Linear Regression
from sklearn.datasets import make_regression
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

X, y = make_regression(
    n_samples=100,
    n_features=1,
    noise=15,
    random_state=0
)

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=0
)

model = LinearRegression()

model.fit(X_train, y_train)

prediction = model.predict(X_test)

print("R2 Score:", r2_score(y_test, prediction))
```

Output

```
... R2 Score: 0.8677570207610111
```

Result

Thus, the Linear Regression algorithm was successfully implemented, and the model accurately predicted continuous values with a high R^2 score.